

Messier: A High-Resolution Corpus for Cross-Benchmark Agent Evaluation

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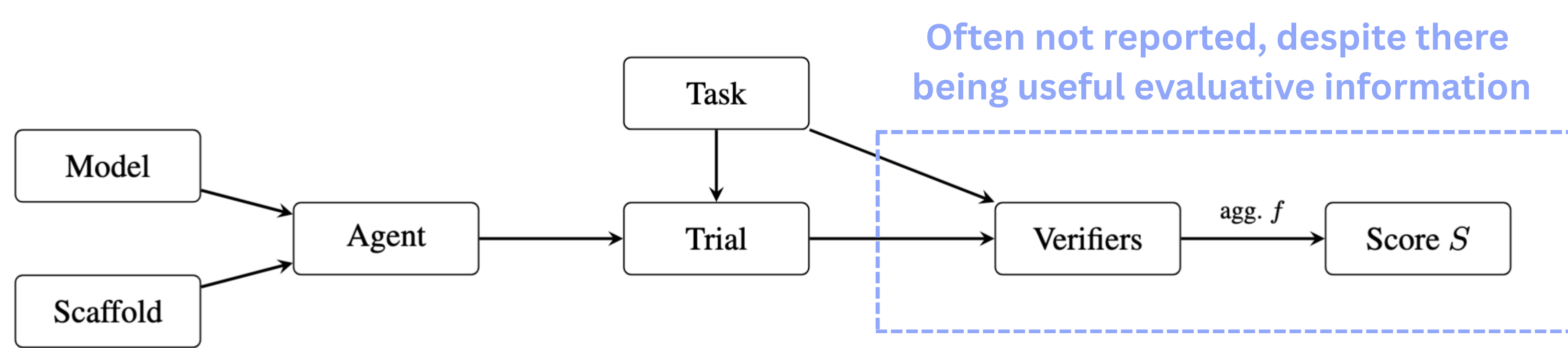
Research question: How can AI agents be compared across different benchmarks?

Contributions

- 1) Consolidated agent evaluation corpus, spanning 31 benchmarks, 714 agents, 12000 tasks, providing macro- and micro-scale views of agent performance: benchmark, environment, task, verifier.
- 2) Capability indices from open data that correspond to Epoch ECI rankings at Spearman's $\rho = 0.84$.
- 3) Evidence against reporting aggregated results, enabled by information at verifier granularity.

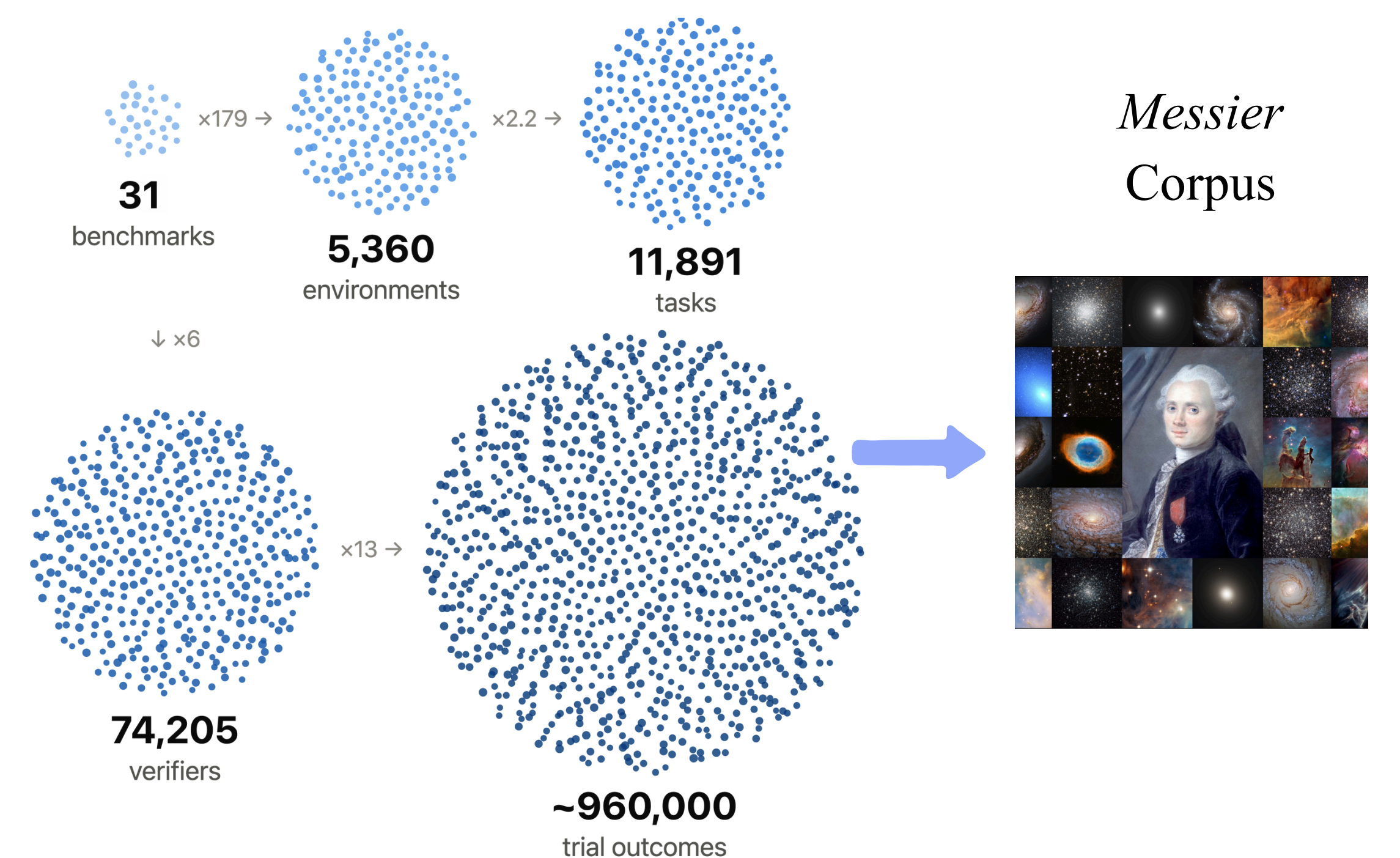
Overview

1) Motivation



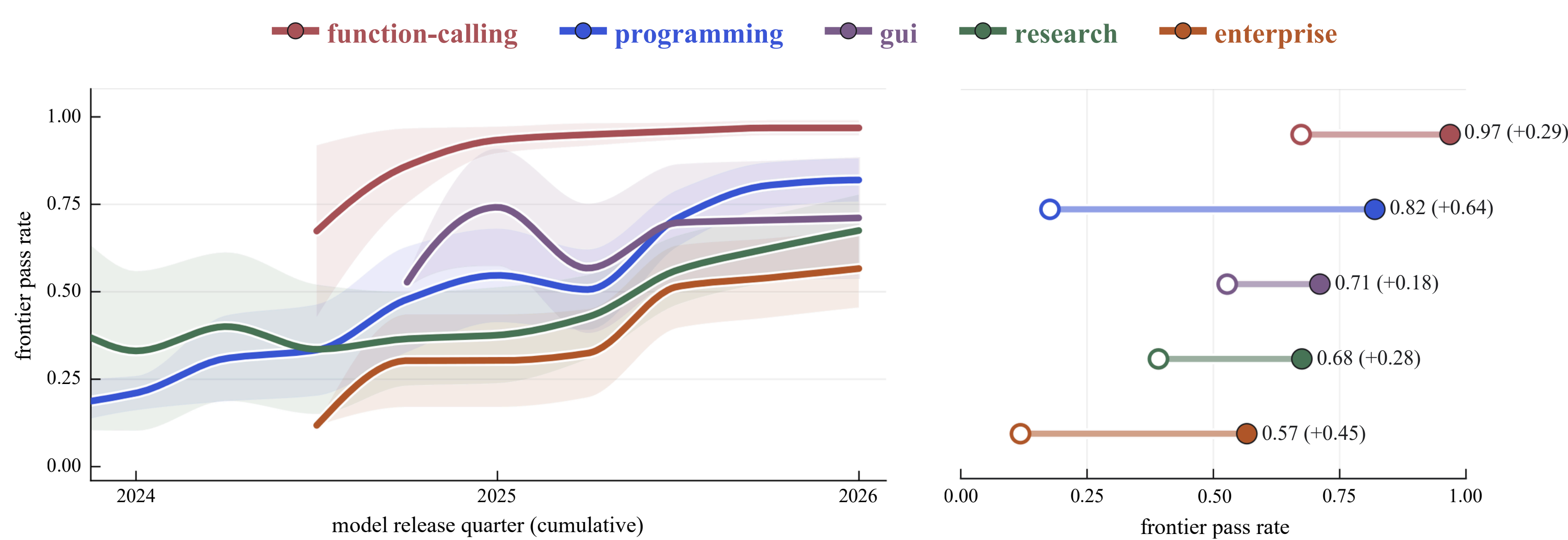
An agent is the composition of a model and a scaffold, a task specifies one or more verifiers. A trial is one execution of an agent on a task. Each verifier produces an outcome, and the function f combines these into a single score S .

2) Scale



Analysis & Applications

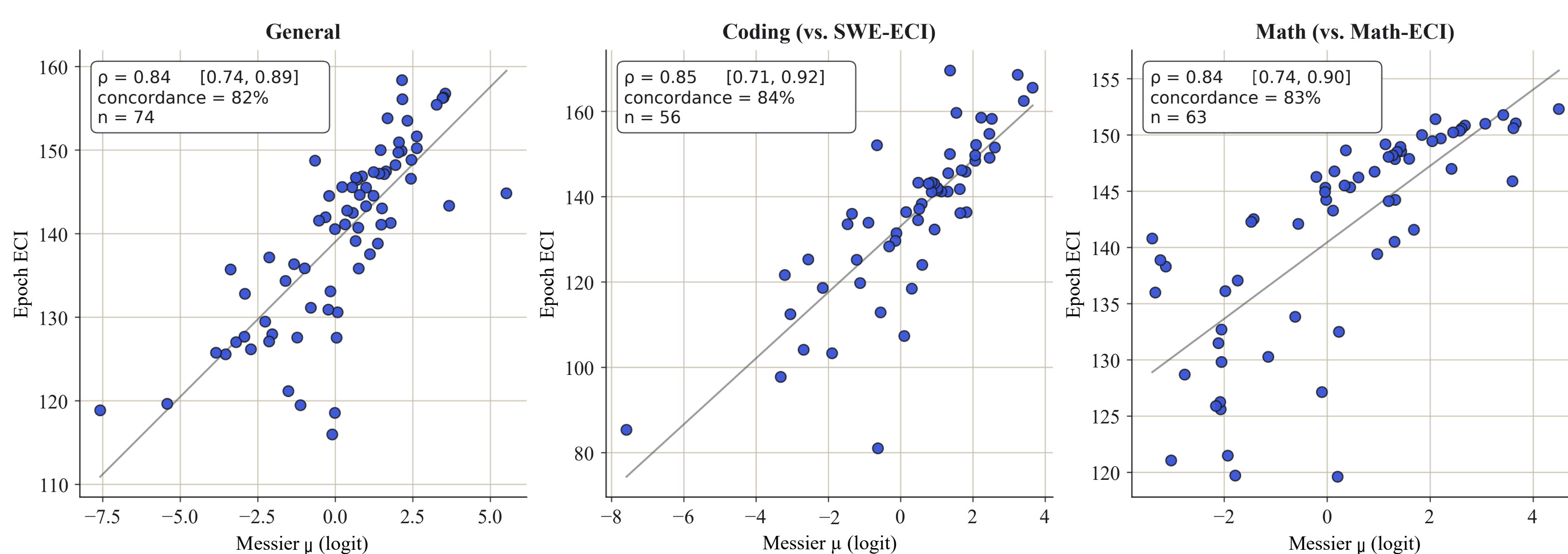
Frontier progress is uneven across benchmark domains



Aggregation can obscure progress

Benchmark	All-pass (original)	Soft mean (fraction)	Majority ($\geq 50\%$)	Rank shift (ρ)
HarveyAI-Lab	0.20%	28.80%	29.80%	0.667
tau-bench	47.90%	75.40%	80.90%	0.857
TheAgentCo.	15.30%	34.00%	35.80%	0.981

Open data IRT scales track Epoch ECI



Task difficulty prediction

Target (β z-scored)	Text Input	RidgeCV Model	Baseline
Within-Benchmark	Task only (task)	$\rho = 0.263$	$\rho = -0.081$
Difficulty	Task + Env + Action Space (all)	$\rho = 0.298$	$\rho = -0.081$

Other ideas

- Train models such as Jev, Laya, CLMs to act as verifiers
- Hierarchical IRT fits, or an alternative to the shared scale
- RAG for environment, task or verifier design

Conclusions

- Messier **makes agent performance comparable across benchmarks** by standardizing agents, tasks, verifiers, and scoring rules.
- **Progress is uneven:** function calling is near saturation, while enterprise workflows remain hard.
- Verifier-level scoring **reveals partial progress and can change agent rankings**.
- Messier **supports custom capability indexing**, enabling fine-grained slicing by SOC/NAICS labor codes, action spaces, environment state, verifiers, pass rate thresholds and more.
- Task metadata **modestly predicts difficulty**, while trajectories **enable future auditing** of evaluation failures and safety risks.

Scan for paper, data and code!



<https://andromede-ai.github.io/messier/>